Ques: Segregate 0s and 1s

M-1

DSA in C++

arr[] = [0, 0, 0, 0, 0, 1, 1, 1, 1, 1]

zeros = ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5

ones = ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5

(Lecture 21)

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Summary Notes

Ques: Segregate 0s and 1s Method-2

[0, 1, 0, 1, 0, 0, 1, 1, 1, 0]

0 0 0 0 0 1 1 1 1 1

i

j

Ques: Segregate 0s and 1s

```
int i = 0, j = arr.size() - 1;
while(i<j){
    if(arr[i] == 0) i++;
    if(arr[j] == 1) j--;
    if(arr[i] == 1 && arr[j] == 0){
        swap(arr[i],arr[j]);
        i++;
        j--;
    }
}
```

1 0 1 0 1 0 0

i

0 0 0 1 0 1 1

j

* The Problem *

Ques: Two Sum

```
          0 1 2  3
nums = [2,7,11,15], target = 22
```

ans = {1, 3}

$$\begin{array}{l} \quad\;\; 0 \quad 1 \quad 2 \\ nums = \{4, 3, 2\},\ tar = 6 \end{array}$$

ans = {0, 2}

2   7   11   15

tar = 22

Leetcode 1

Ques: Missing in Array

PW Earners

arr = {6, 4, 1, 5, 3, 0}

missing no = 2

M-1 : Nested Loops

M-2 : Sorting

0 1 2 3 4 5
{0, 1, 3, 4, 5, 6}

n=6

```
for(i = 0 to n) {
    bool flag = false
    for(int ele : arr)
        if(i==ele)
            flag = true
            break
        }
    }
} if flag==false → return
```

Leetcode 268

Method 3 ⇒ Extra Space

Ques: Missing in Array

arr = { 5, 4, 1, 6, 0, 2 }

missing no = 3

n = 6

TC = O(n)

AS = O(n)

flag =

0	1	2	3	4	5	6
T	T	T	F	T	T	T

Leetcode 268

Method-4 : Maths

Ques: Missing in Array

arr = { 5, 4, 1, 6, 0, 2 }

missing no = 3

n = 6

Code → HW

arraySum = 18

zeroToNSum = $\frac{n(n+1)}{2} = \frac{6 \cdot 7}{2} = 21$

TC = O(n)

21 - 18 = 3

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Ques: Wave Array

Ques: 66. Plus One

arr = {9, 9, 9}

ans = {1, 0, 0, 0}

$$\begin{array}{ccc} 1 & 1 & \\ 9 & 9 & 9 \\ & + & 1 \\ \hline 0 & 0 & 0 \\ \hline \end{array}$$

carry = $\cancel{1}$ 1 $\cancel{1}$ 1

ans = {0, 0, 0, 1}

↓

rev

Ques: 66. Plus One

arr = {1, 2, 9, 9}

ans = {1, 3, 0, 0}

$$\begin{array}{r} 0\,1\;\;1\;\; \\ 1\;2\;9\;9 \\ +\;1 \\ \hline 1\;3\;0\;0 \\ \hline \end{array}$$

carry = $\cancel{1}\,\cancel{1}$
$\cancel{1}\,\cancel{0}\,0$

---

arr = {9, 1, 0, 9}

ans = {9, 1, 1, 0}

ans = {0, 0, 3, 1}

↓

reverse